

5.1 Determinants

- The determinant of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $ad - bc$. Singular matrix $A = \begin{bmatrix} a & xa \\ c & xc \end{bmatrix}$ has $\det = 0$.
- **Row exchange reverse signs** $PA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$ has $\det PA = bc - ad = -\det A$.
- The determinant of $\begin{bmatrix} xa + yA & xb + yB \\ c & d \end{bmatrix}$ is $x(ad - bc) + y(Ad - Bc)$. **Det is line in row 1 by itself**

- **Elimination** $EA = \begin{bmatrix} a & b \\ 0 & d - \frac{c}{a}b \end{bmatrix}$ $\det EA = a \left(d - \frac{c}{a}b \right) = \text{product of pivots} = \det A$.
- If A is n by n then 1, 2, 3, 4 remain true: **$\det = 0$ when A is singular**, $\det$ reverses sign when rows are exchanged, $\det$ is **linear in row 1** by itself, $\det =$ **product of the pivots**. Always $\det BA = (\det B)(\det A)$ and $\det A^T = \det A$.
This is an amazing number.

The Properties of Determinants

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ has inverse } A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (1)$$

- 1. The determinant of the n by n identity matrix is 1.**

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \quad \text{and} \quad \begin{vmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{vmatrix} = 1$$

- 2. The determinant changes sign when two rows are exchanged (sign reversal):**

$$\text{check: } \begin{vmatrix} c & d \\ a & b \end{vmatrix} = - \begin{vmatrix} a & b \\ c & d \end{vmatrix} \quad (\text{both sides equal } bc-ad)$$

3.The determinant is a linear function of each row separately

multiply row 1 by any number t
det is multiplied by t

$$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

add row 1 of A to row 1 of A' :
then determinants add

$$\begin{vmatrix} a + a' & b + b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$$

$$A = \begin{vmatrix} 4 & 8 & 8 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 4 \begin{vmatrix} 1 & 2 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} \quad \text{and} \quad \begin{vmatrix} 4 & 8 & 8 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 4 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 0 & 8 & 8 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

4.If two rows of A are equal, then $\det A = 0$.

Equal rows Check 2 by 2: $\begin{vmatrix} a & b \\ a & b \end{vmatrix} = 0$

5. Subtracting a multiple of one row from another row leaves $\det A$ unchanged.

l time row 1
from row 2

$$\begin{vmatrix} a & b \\ c - la & d - lb \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

6. A matrix with a row of zeros has $\det A = 0$.

Row of zeros

$$\begin{vmatrix} 0 & 0 \\ c & d \end{vmatrix} = 0 \text{ and } \begin{vmatrix} a & b \\ 0 & 0 \end{vmatrix} = 0$$

7. If A is triangular then $\det A =$ product of diagonal entries.

Triangular

$$\begin{vmatrix} a & b \\ 0 & d \end{vmatrix} = ad \text{ and also } \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} = ad$$

Diagonal matrix $\det \begin{bmatrix} a_{11} & & & 0 \\ & a_{22} & & \\ & & \ddots & \\ 0 & & & a_{nn} \end{bmatrix} = (a_{11})(a_{22}) \dots \dots (a_{nn}).$

8. If A is singular then $\det A = 0$. If A is invertible then $\det A \neq 0$.

Singular $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is singular if and only if $ad - bc = 0$.

9. The determinant of AB is $\det A$ times $\det B$: $|AB| = |A||B|$,

Product rule $\begin{vmatrix} a & b \\ c & d \end{vmatrix} \begin{vmatrix} p & q \\ r & s \end{vmatrix} = \begin{vmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{vmatrix}$

10. The transpose A^T has the same determinant as A .

Transpose $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & c \\ b & d \end{vmatrix}$ since both sides equal $ad - bc$

5.2 Permutations and Cofactors

- 2 by 2: ad-ac has $2!$ Terms with $\pm$ signs. n by n : $\det A$ add $n!$ terms with $\pm$ signs.
- For $n = 3$, $\det A$ adds $3! = 6$ terms. Two terms are $+a_{12}a_{23}a_{31}$ and $-a_{13}a_{22}a_{31}$. Row 1,2,3 and columns 1,2,3 appear once in each term.
- That minus sign came because the column order 3,2,1 needs one exchange to recover 1,2,3
- The six terms include
$$+a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) = a_{11}(\text{cofactor } C_{11})$$
Always $\det A = a_{11}C_{11} + a_{12}C_{12} + \dots + a_{1n}C_{1n}$. Cofactors are determinants of size $n - 1$.

$$A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

$$\det A = 2 \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} + 1 \begin{bmatrix} -1 & -1 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} + 0 + 0$$

$$= 2 \left\{ 2 \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} + 1 \begin{vmatrix} -1 & -1 \\ 0 & 2 \end{vmatrix} \right\} + 1 \left\{ -1 \begin{vmatrix} 2 & -1 \\ 0 & 2 \end{vmatrix} + 1 \begin{vmatrix} 0 & -1 \\ 0 & 2 \end{vmatrix} \right\}$$

$$= 2 \{ 2(4 - 1) + (-2 + 0) \} + \{ (-1)(4 - 1) + 0 \}$$

$$= 2 \{ 2(3) - 2 \} - 3$$

$$= 5$$

The Pivot Formula

$A = LU$, the pivots $d_1, \dots, d_n$ are on the diagonal of the upper triangular U

$$\det A = (\det L)(\det U) = (1)(d_1 d_2 \dots d_n)$$

$$(\det P)(\det A) = (\det L)(\det U) \text{ gives } \det A = \pm(d_1 d_2 \dots d_n)$$

Example

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad PA = \begin{bmatrix} 4 & 5 & 6 \\ 0 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \quad \det A = -(4)(2)(1) = -8$$

Example

$$\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & \ddots & \\ & & \ddots & \ddots & -1 \\ & & & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & & & & \\ -\frac{1}{2} & 1 & & & \\ & -\frac{2}{3} & 1 & & \\ & & \ddots & \ddots & \\ & & & \frac{n-1}{n} & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & & & \\ & \frac{3}{2} & -1 & & \\ & & \frac{4}{3} & -1 & \\ & & & \ddots & \ddots \\ & & & & \frac{n+1}{n} \end{bmatrix}$$

Pivots from determinants

The k th pivot is $d_k = \frac{d_1 d_2 \dots d_k}{d_1 d_2 \dots d_{k-1}} = \frac{\det A_k}{\det A_{k-1}}$

The Big Formula for Determinant

**3 by 3
determinant**

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) \\ + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

Determinant by Cofactors

$$A = \begin{bmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{bmatrix} \quad \text{signs}(-1)^{i+j} = \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

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